

Monte Carlo Study of the 2D Ising Model: Finite-Size Scaling and Critical Temperature Estimation

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The two-dimensional Ising model provides a minimal setting for studying phase transitions and critical phenomena. In this report we investigate the equilibrium properties of the 2D Ising model on square lattices with periodic boundary conditions using single-spin Metropolis updates. For a 20×20 lattice we study equilibration from ordered and disordered initial states, showing that the burn-in time increases when approaching the critical region: a few 10^2 sweeps suffice at $T = 1.0 J/k_B$, whereas of order 10^4 – 10^5 sweeps are required at $T = 2.4 J/k_B$. Using long runs at fixed temperature we construct normalised histograms of the energy per spin, finding a narrow distribution deep in the ordered phase and a much broader distribution with enhanced fluctuations near criticality. We then employ OpenMP to parallelise independent simulations at different temperatures, achieving a speed-up factor $S \approx 3.4$ for five temperatures at fixed $L = 20$. Finally, we map out the temperature dependence of $\langle \epsilon \rangle$, $\langle |m| \rangle$, C_V/N and χ/N for different lattice sizes in the range $2.1 \leq T/(J/k_B) \leq 2.4$. From the peaks of the heat capacity and susceptibility we extract pseudo-critical temperatures $T_c(L)$ and perform a finite-size scaling analysis based on $T_c(L) = T_c(\infty) + a/L$. A weighted linear fit of $T_c(L)$ versus $1/L$ yields $T_c(\infty) = 2.25 \pm 0.02 J/k_B$, in good agreement with Onsager's exact result $T_c = 2.269 J/k_B$.

I. INTRODUCTION

Motivation and Objectives

The Ising model provides a simple yet powerful framework to study cooperative phenomena in statistical mechanics. In this report, we focus on the two-dimensional version of the model, which captures essential aspects of ferromagnetic behaviour. By simulating the system at different temperatures, we aim to explore how macroscopic properties such as magnetisation and energy emerge from microscopic spin interactions. We will investigate how the equilibrium properties of the system depend on temperature and lattice size, analysing quantities such as the mean energy, mean magnetisation, heat capacity, and magnetic susceptibility. Finally, by studying finite lattice sizes, we will extrapolate our results to the infinite-lattice limit and compare the estimated critical temperature with the exact analytical result derived by Onsager in 1944 [1].

The remainder of this report is organised as follows. In Sec. II we introduce the numerical methods used throughout the work, including the analytical solution of the 2×2 lattice, the Metropolis Markov chain Monte Carlo scheme, the treatment of periodic boundary conditions, and the finite-size scaling framework. Sec. III presents our numerical results and discussion: we first validate the code on the 2×2 system and study equilibration for $L = 20$, then investigate the full energy distribution, assess the performance of the OpenMP-parallel implementation, and finally analyse the critical region and extract $T_c(L = \infty)$ from finite-size data. All code developed for this report is available via this GitHub repository ¹.

Definitions, Notation and Units

We consider a two-dimensional square lattice of spins s_i , where each spin can take one of two possible states, $s_i = \pm 1$. A lattice of side length L contains $N = L^2$ spins in total. A spin configuration, or microstate, represents the state of all spins in the system and can be written as a vector

$$\mathbf{s} = (s_1, s_2, \dots, s_N),$$

which in numerical implementations is represented as an $L \times L$ matrix.

The total energy of a given configuration is defined as

$$E(\mathbf{s}) = -J \sum_{\langle kl \rangle} s_k s_l, \quad (1)$$

where the sum runs over all neighbouring pairs of spins, without double counting, and J is the coupling constant that determines the interaction strength [2]. Periodic boundary conditions are applied, so that each spin interacts with its four nearest neighbours (left, right, up, and down).

The total magnetisation of the system is given by [2]

$$M(\mathbf{s}) = \sum_{i=1}^N s_i. \quad (2)$$

To facilitate comparison between systems of different sizes, we use the energy and magnetisation per spin:

$$\epsilon(\mathbf{s}) = \frac{E(\mathbf{s})}{N}, \quad m(\mathbf{s}) = \frac{M(\mathbf{s})}{N}.$$

At temperature T , the probability of the system being in configuration $\mathbf{s}$ follows the Boltzmann distribution:

$$p(\mathbf{s}; T) = \frac{1}{Z} e^{-\beta E(\mathbf{s})}, \quad (3)$$

where Z is the partition function,

$$Z = \sum_{\text{all } \mathbf{s}} e^{-\beta E(\mathbf{s})}, \quad (4)$$

and $\beta = \frac{1}{k_B T}$ is the inverse temperature, with k_B the Boltzmann constant [2].

Since we are working with dimensionless spins, the coupling constant J sets the unit of energy. Thus, the natural choice of units is

$$[E] = J, \quad [T] = \frac{J}{k_B}.$$

All other quantities are expressed consistently in these units.

¹ <https://github.com/teresm/Comp.Phys.Project/tree/master/P4>

II. METHODS

2×2 lattice

To establish a baseline for our simulations, we first enumerate all microstates of a 2×2 Ising lattice with Periodic Boundary Conditions (PBC). Spins take values $s_i \in \{-1, +1\}$ and interact only with their four nearest neighbours (left, right, up, down). We use the standard Ising Hamiltonian without external field from eq. 1 and, for the 2×2 case with PBC, we *allow each neighbour pair to be counted twice*. This makes the energy expression operationally identical to summing over every site and its four neighbours without the usual $1/2$ correction factor. With this convention the 2×2 lattice has 16 interaction terms in total. The total magnetisation can be calculated from eq. 2 using $N = 4$.

By grouping microstates according to (i) the number of spins in the $+1$ state n_+ , (ii) the total energy E , and (iii) the total magnetisation M , we obtain the complete classification shown in Table I. Energies are reported in units of J .

TABLE I. 2×2 lattice with PBC: exhaustive grouping of microstates by number of $+1$ spins (n_+), total energy E , total magnetisation M , and degeneracy g . Energies use the double-counted-neighbour convention described in the text.

n_+	E	M	g
0	$-16J$	-4	1
1	0	-2	4
2	0	0	4
2	$+16J$	0	2
3	0	$+2$	4
4	$-16J$	$+4$	1

The two ferromagnetic macrostates (all spins aligned) yield $E = -16J$ and $|M| = 4$, while the two antiferromagnetic “checker-board” configurations produce $E = +16J$ and $M = 0$. All remaining mixed configurations have $E = 0$ and $M \in \{-2, 0, +2\}$ with the degeneracies listed above. This tabulation will be used in later sections as an analytical reference to validate the C++ implementation and the sampling procedures at finite temperature.

Using the Boltzmann weight from eq. 3 with the energy $E(\mathbf{s}) = -J \sum_{\langle kl \rangle} s_k s_l$ and the double-counting convention for nearest neighbours, the 2×2 lattice ($N = 4$ spins) yields three energy levels: $E = -16J$ (degeneracy 2), $E = 0$ (degeneracy 12), and $E = +16J$ (degeneracy 2). Writing $A = 16\beta J$ and $D = 3 + \cosh A$, the partition function is

$$Z = \sum_{\mathbf{s}} e^{-\beta E(\mathbf{s})} = 2e^A + 12 + 2e^{-A} = 4[3 + \cosh A] = 4D. \quad (5)$$

From the above spectrum we obtain

$$\langle E \rangle = \frac{1}{Z} \sum_{\mathbf{s}} E e^{-\beta E(\mathbf{s})} = -\frac{64J \sinh A}{Z} \quad (6)$$

$$\langle E^2 \rangle = \frac{1}{Z} \sum_{\mathbf{s}} E^2 e^{-\beta E(\mathbf{s})} = \frac{1024J^2 \cosh A}{Z}. \quad (7)$$

In terms of energy per spin $\langle \epsilon \rangle = \langle E \rangle / N$ this gives

$$\langle \epsilon \rangle = -\frac{4J \sinh A}{D}, \quad \langle \epsilon^2 \rangle = \frac{16J^2 \cosh A}{D}. \quad (8)$$

Grouping states by total magnetisation M (with the replacement $M \rightarrow |M|$), the nonzero contributions are: $|M| = 4$ with $E = -16J$ (degeneracy 2) and $|M| = 2$ with $E = 0$ (degeneracy 8). Hence

$$\langle |M| \rangle = \frac{8e^A + 16}{Z}, \quad \langle M^2 \rangle = \frac{32e^A + 32}{Z}. \quad (9)$$

For $\langle m \rangle = \langle M \rangle / N$ this implies

$$\langle |m| \rangle = \frac{e^A + 2}{2D}, \quad \langle m^2 \rangle = \frac{e^A + 1}{2D}. \quad (10)$$

We can obtain now the normalised heat capacity and susceptibility expressions.

$$\frac{C_V(T)}{N} = N \frac{1}{k_B T^2} (\langle \epsilon^2 \rangle - \langle \epsilon \rangle^2) \quad (11)$$

$$\frac{C_V(T)}{N} = \beta^2 k_B 64J^2 \left[\frac{\cosh A}{D} - \frac{\sinh^2 A}{D^2} \right], \quad (12)$$

and, with the $M \rightarrow |M|$ convention,

$$\frac{\chi(T)}{N} = N \frac{1}{k_B T} (\langle m^2 \rangle - \langle |m| \rangle^2) \quad (13)$$

$$\frac{\chi(T)}{N} = \beta \left[\frac{2(e^A + 1)}{D} - \frac{(e^A + 2)^2}{D^2} \right]. \quad (14)$$

These closed-form expressions will serve as analytical benchmarks to validate our C++ implementation.

Single-spin energy increments

We consider the standard 2D Ising Hamiltonian with nearest-neighbor interactions and no external field,

$$\mathcal{H}(\{s\}) = -J \sum_{\langle ij \rangle} s_i s_j, \quad s_i \in \{-1, +1\},$$

with periodic boundary conditions (so every spin has four neighbours). Flipping a single spin $s_k \rightarrow -s_k$ changes only the four bonds attached to k . Denoting by $\mathcal{N}(k)$ the set of its nearest neighbors and by $n_k = \sum_{j \in \mathcal{N}(k)} s_j$ the local neighbor sum, the energy increment is

$$\begin{aligned} \Delta E &\equiv E_{\text{after}} - E_{\text{before}} = \\ &= -J [\dots + (-s_k s_1 - s_k s_2 - s_k s_3 - s_k s_4) + \dots] \\ &\quad + J [\dots + (s_k s_1 + s_k s_2 + s_k s_3 + s_k s_4) + \dots] \\ \Delta E &= 2J s_k \sum_{j \in \mathcal{N}(k)} s_j = 2J s_k n_k. \end{aligned} \quad (15)$$

Because $n_k \in \{-4, -2, 0, 2, 4\}$ on a square lattice, the product $s_k n_k$ also takes values in $\{-4, -2, 0, 2, 4\}$. Hence, for any lattice of size $L > 2$, there are only five possible energy increments:

$$\Delta E \in \{-8J, -4J, 0, 4J, 8J\}.$$

Since ΔE can take only five values, we precompute a small table of Boltzmann factors ($e^{-\beta \Delta E}$) that can be used directly in the code:

$$\mathcal{B} = \{e^{\beta(8J)}, e^{\beta(4J)}, 1, e^{-\beta(4J)}, e^{-\beta(8J)}\}.$$

Periodic boundary conditions

Periodic boundary conditions (PBC) are applied so that every spin has exactly four neighbours, even at the edges of the lattice. Instead of using conditional if-tests to handle the borders inside the Monte Carlo loop, which can slow down the execution, we implemented the wrapping of indices using the modulus operator ($\text{mod } L$). This approach is simple, branch-free, and allows the compiler to optimise the loop efficiently.

In practice, when computing the neighbour spins of a given site (i, j) , the indices of the right, left, up and down neighbours are obtained as:

$$\begin{aligned} (i+1) \bmod L, \quad (i-1+L) \bmod L, \\ (j+1) \bmod L, \quad (j-1+L) \bmod L. \end{aligned}$$

Algorithm 1 Periodic boundary conditions for a 2D lattice

Input: Lattice $s[i][j]$ of size $L \times L$

Output: Neighbour sum n_{ij} for each site

for $i = 0$ to $L - 1$ **do**

for $j = 0$ to $L - 1$ **do**

$i_{\text{up}} \leftarrow (i + 1) \bmod L$

$i_{\text{down}} \leftarrow (i - 1 + L) \bmod L$

$j_{\text{right}} \leftarrow (j + 1) \bmod L$

$j_{\text{left}} \leftarrow (j - 1 + L) \bmod L$

$n_{ij} \leftarrow s[i_{\text{up}}][j] + s[i_{\text{down}}][j] + s[i][j_{\text{right}}] + s[i][j_{\text{left}}]$

This guarantees that sites on one edge of the lattice interact correctly with those on the opposite edge. The following pseudocode summarises the algorithm used to evaluate neighbour interactions under periodic boundary conditions.

This method avoids explicit boundary checks and ensures smooth and efficient updates during the Metropolis algorithm. It is general for any lattice size L and introduces negligible computational overhead compared to using conditionals.

Markov chain Monte Carlo (Metropolis)

We sample spin configurations at fixed temperature T using single-spin Metropolis updates (see Algorithm II). Starting from an initial lattice $s_{ij} \in \{-1, +1\}$, one Monte Carlo sweep (MCS) attempts $N = L^2$ random single-spin flips. For a proposed flip $s_{ij} \rightarrow -s_{ij}$, the local energy increment is

$$\Delta E = 2J s_{ij} (s_{i,j+1} + s_{i,j-1} + s_{i+1,j} + s_{i-1,j}), \quad (16)$$

with periodic boundaries. The move is accepted with probability $p = \min\{1, e^{-\beta \Delta E}\}$, where $\beta = 1/(k_B T)$. Since $\Delta E \in \{-8J, -4J, 0, 4J, 8J\}$ on a square lattice, we precompute once per temperature the five Boltzmann factors $\{e^{-\beta \Delta E}\}$ and reuse them during the sweep. (For the special case $L = 2$ under periodic boundaries, every bond is counted twice; we implement this by doubling ΔE or, equivalently, by precomputing the set $\{-16J, -8J, 0, 8J, 16J\}$.) After a burn-in stage where no measurements are recorded, we collect measurements every few sweeps to reduce autocorrelation and then form ensemble averages:

$$\begin{aligned} \langle \epsilon \rangle &= \frac{\langle E \rangle}{N}, & \langle |m| \rangle &= \frac{\langle |M| \rangle}{N}, \\ \frac{C_V}{N} &= \frac{\beta^2}{N} (\langle E^2 \rangle - \langle E \rangle^2), \\ \frac{\chi}{N} &= \frac{\beta}{N} (\langle M^2 \rangle - \langle M \rangle^2). \end{aligned}$$

Finite-size scaling and critical behaviour

In the thermodynamic limit for an infinite lattice ($L \rightarrow \infty$), the 2D Ising model exhibits a phase transition between a ferromagnetic phase with spontaneous magnetisation and a paramagnetic phase with $\langle m \rangle = 0$. The exact critical temperature at zero external field was obtained by Onsager and is given by [1]:

$$T_c(L = \infty) = \frac{2J}{k_B \ln(1 + \sqrt{2})} \approx 2.269 \frac{J}{k_B}. \quad (17)$$

The behaviour of thermodynamic observables is described by power laws characterised by critical exponents. For the infinite 2D Ising

Algorithm 2 Metropolis MCMC for the 2D Ising model at fixed temperature

Input: lattice size L , temperature T , coupling J , burn-in B , total sweeps S , sampling interval k

Initialise: spins $s_{ij} \in \{\pm 1\}$, compute (E, M) ,
set $\beta \leftarrow 1/T$, $N \leftarrow L^2$

Precompute Boltzmann table $\mathcal{B} = \{e^{-\beta \Delta E}\}$
for $\Delta E \in \{-8J, -4J, 0, 4J, 8J\}$
(if $L = 2$: use $\{-16J, -8J, 0, 8J, 16J\}$ instead)

for $\text{sweep} = 1$ to S **do**

for $t = 1$ to N **do** ▷ one Monte Carlo sweep

 pick a random site (i, j)

$n = s_{i,j+1} + s_{i,j-1} + s_{i+1,j} + s_{i-1,j}$

$\Delta E \leftarrow 2J s_{ij} n$ (if $L = 2$: $\Delta E \leftarrow 2\Delta E$)

if $\Delta E \leq 0$ **or** $u < e^{-\beta \Delta E}$ with $u \sim \mathcal{U}[0, 1)$ **then**
 $s_{ij} \leftarrow -s_{ij}; \quad M \leftarrow M - 2s_{ij}^{\text{old}}; \quad E \leftarrow E + \Delta E$

if $\text{sweep} > B$ **and** $(\text{sweep} - B) \bmod k = 0$ **then**
 record $E, M, |M|, E^2, M^2$

compute $\langle \epsilon \rangle, \langle |m| \rangle, C_V/N, \chi/N$ from recorded samples

model one has

$$\langle |m| \rangle \propto |T - T_c(L = \infty)|^\beta, \quad (18)$$

$$\frac{C_V}{N} \propto |T - T_c(L = \infty)|^{-\alpha}, \quad (19)$$

$$\frac{\chi}{N} \propto |T - T_c(L = \infty)|^{-\gamma}, \quad (20)$$

for temperatures T close to $T_c(L = \infty)$, where β , α and γ are the magnetisation, specific-heat and susceptibility exponents, respectively [2]. For the 2D Ising universality class these are known exactly as

$$\beta = \frac{1}{8}, \quad \alpha = 0, \quad \gamma = \frac{7}{4}. \quad (21)$$

The divergence of fluctuations is associated with a diverging correlation length ξ , which measures the typical size of spin clusters,

$$\xi \propto |T - T_c(L = \infty)|^{-\nu}, \quad (22)$$

with correlation-length exponent $\nu = 1$ for the 2D Ising model [2]. In an infinite system ξ can grow without bound, but in a finite system of linear size L the largest possible correlation length is limited by the system size, $\xi \lesssim L$. This finite-size cutoff smooths the singularities of the thermodynamic limit into rounded peaks whose position and height depend on L .

A simple way to connect the finite and infinite systems is to associate the pseudo-critical temperature $T_c(L)$ of a finite lattice with the condition that the correlation length becomes comparable to the system size,

$$\xi(T_c(L)) \sim L. \quad (23)$$

Inserting the scaling form (22) then yields

$$|T_c(L) - T_c(L = \infty)| \propto L^{-1/\nu}. \quad (24)$$

For the 2D Ising case with $\nu = 1$ this reduces to the linear scaling relation

$$T_c(L) - T_c(L = \infty) = aL^{-1}, \quad (25)$$

where a is a constant. Equation (25) implies that a plot of $T_c(L)$ versus $1/L$ should fall approximately on a straight line, whose intercept at $1/L \rightarrow 0$ provides an estimate of the infinite-volume critical

temperature $T_c(L = \infty)$ [3]. In Sec. III we will use our numerical estimates of $T_c(L)$ from the peak positions of C_V/N and χ/N to perform such an extrapolation and compare the result to Onsager's exact value.

The same finite-size scaling arguments also predict how observables behave at (or very close to) $T_c(L)$ as the lattice size is varied [2]. Replacing $|T - T_c|$ in the infinite-system power laws by $L^{-1/\nu}$ leads to

$$\langle |m| \rangle \propto L^{-\beta/\nu}, \quad (26)$$

$$\frac{C_V}{N} \propto L^{\alpha/\nu}, \quad (27)$$

$$\frac{\chi}{N} \propto L^{\gamma/\nu}. \quad (28)$$

Tools

All simulations were implemented in C++. The core Ising model code consists of a single-spin Metropolis update routine, written to make heavy use of inlined functions and precomputed Boltzmann factors in order to minimise overhead in the innermost loops. Periodic boundary conditions are handled via modular arithmetic on lattice indices, which avoids conditional branches and facilitates compiler optimisation.

To accelerate parameter scans, in particular the exploration of several temperatures for a fixed lattice size, we used OpenMP for shared-memory parallelisation. Independent Markov chains at different temperatures were executed concurrently by annotating the outer temperature loop with a `#pragma omp parallel` for directive.

Post-processing and visualisation were performed in Python using NumPy for data handling and Matplotlib for generating the plots of equilibration, histograms, thermodynamic observables, and finite-size scaling fits. In addition, the large language model ChatGPT was used as a writing aid to help draft and refine the text of this report (organisation of sections, phrasing of explanations, and editing of LaTeX), but not to generate or modify the numerical results themselves.

III. RESULTS AND DISCUSSION

2x2 lattice and $T = 1 \frac{J}{k_B}$

We first validated our implementation on the 2×2 lattice with periodic boundary conditions (PBC) at fixed temperature $T = 1 \frac{J}{k_B}$, varying the number of Monte Carlo sweeps (one sweep $\equiv L^2$ single-spin flip attempts). Table II reports the measured observables and the number of accepted flips. Analytical benchmarks for $L = 2$ with PBC and double counting (Sec. II, Eqs. (8), (10), (12), (14)) at $T = 1$ are:

$$\langle \epsilon \rangle = -3.9999973 \text{ J}, \quad \langle |m| \rangle = 0.99999955,$$

$$\frac{C_V}{N} = 4.3213 \times 10^{-5}, \quad \frac{\chi}{N} = 1.3504 \times 10^{-6} \text{ J}^{-1}.$$

The MC estimates for the *means* (energy per spin and absolute magnetisation per spin) match the closed-form predictions within printing precision, which confirms that the acceptance rule, PBC, and energy convention (double counting in 2×2) are consistent. In contrast, the *fluctuation* observables (C_V/N and χ/N) appear as zeros, while the analytical values are small but nonzero. This is expected for a 2×2 lattice at $T = 1 \text{ J/k}_B$: the energy gap for creating a defect is $\Delta E = 16 \text{ J}$, hence the acceptance probability for an uphill move is $e^{-\beta \Delta E} = e^{-16} \approx 1.1 \times 10^{-7}$. With $N = 4$ sites, one sweep attempts 4 flips, so the expected number of accepted uphill flips per sweep is $\sim 4 \times e^{-16} \approx 4.4 \times 10^{-7}$. Consequently,

TABLE II. MC results for $L = 2$, $T = 1 \text{ J/k}_B$ as a function of the number of Monte Carlo sweeps (S).

S	$\langle \epsilon \rangle [\text{J}]$	$\langle m \rangle$	C_V/N	$\chi/N [\text{J}^{-1}]$	Accepted flips
20 000	-4.0	1.0	0.0	0.0	2
30 000	-4.0	1.0	0.0	0.0	9
40 000	-4.0	1.0	0.0	0.0	4
50 000	-4.0	1.0	0.0	0.0	32
100 000	-4.0	1.0	0.0	0.0	5

the chain spends essentially all its time in the two ferromagnetic ground states, so $\langle E^2 \rangle \simeq \langle E \rangle^2$ and $\langle M^2 \rangle \simeq \langle |M| \rangle^2$, driving C_V/N and χ/N to numerical zero when printed with standard precision.

The counts in Table II show that the number of accepted flips is small and noisy, and *not* strictly proportional to S . This is normal for a process dominated by extremely rare events. Occasionally, once a rare excitation occurs, subsequent flips with lower or negative ΔE can be accepted with probability close to one, producing short bursts (hence the variability, e.g., 32 accepted at $S = 50\,000$).

For such a tiny lattice at low temperature, the mean observables validate the implementation, but fluctuation observables require either (i) higher temperatures, or (ii) vastly longer runs. For the remainder of the study, larger L and temperature, we therefore expect C_V/N and χ/N to become measurable and informative.

Equilibration and burn-in analysis for $L = 20$

To monitor equilibration, we recorded the instantaneous energy per spin $\epsilon = E/N$ after each Monte Carlo cycle and computed its cumulative mean $\langle \epsilon \rangle$ as the simulation progressed. Both ordered (all spins up) and disordered (random) initial states were used to test convergence towards equilibrium. The resulting time series for $T = 1.0 \text{ J/k}_B$ and $T = 2.4 \text{ J/k}_B$ are displayed in Fig. 1, showing both the instantaneous and averaged trajectories.

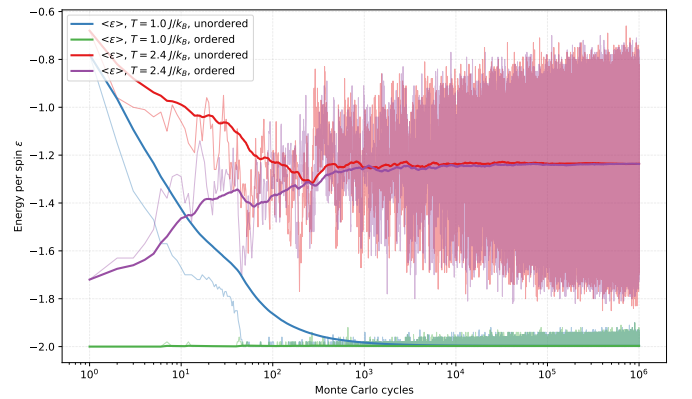


FIG. 1. Evolution of the energy per spin ϵ and its running average $\langle \epsilon \rangle$ for a 20×20 lattice at two temperatures. Thin lines correspond to instantaneous ϵ values, while thick lines show the cumulative average for ordered and unordered initial configurations. A logarithmic x -axis highlights the early transient region.

At $T = 1.0 \text{ J/k}_B$ both simulations—starting from ordered and unordered configurations—stabilise at the same plateau. The system settles into the ferromagnetic state, and fluctuations around the mean are very small. This indicates that the system is already equilibrated after only a few tens of Monte Carlo cycles, making

the required burn-in time negligible at such low temperature (less than 10^2 Monte Carlo cycles if we observe the thin line).

In contrast, at $T = 2.4 \text{ J/k}_B$ the curves exhibit a slower and more irregular approach to the common equilibrium value. The unordered start shows a clear relaxation trend before reaching the steady regime, while the ordered configuration gradually loses magnetisation and drifts toward the same mean energy. The convergence of the two cumulative averages occurs only after $\approx 10^5$ Monte Carlo cycles. Here is when the $\langle \epsilon \rangle$ curves for the unordered and ordered cases reach the same value.

From these observations, we infer that the burn-in time grows rapidly with temperature. For $T = 1.0 \text{ J/k}_B$ it can be safely set to a few hundred sweeps, whereas for $T = 2.4 \text{ J/k}_B$ a burn-in of the order of 10^4 – 10^5 sweeps is necessary to ensure that measurements are independent of the initial condition.

Approximation to energy distribution $p_\epsilon(\epsilon; T)$

To move beyond expectation values, we estimated the full probability distribution of the energy per spin $\epsilon = E/N$ from long Metropolis runs on a 20×20 lattice. For each temperature we performed 10^6 Monte Carlo cycles (MCC), discarding an initial burn-in segment to suppress memory of the starting configuration. Specifically, we used 100 MCC as burn-in at $T = 1.0 \text{ J/k}_B$ and 10^5 MCC at $T = 2.4 \text{ J/k}_B$, reflecting the much slower equilibration. The histograms were constructed from the instantaneous ϵ samples collected after burn-in, normalised as probabilities (bin counts divided by the total number of post-burn-in samples). The resulting probability mass functions are shown in Figs. 2 and 3.

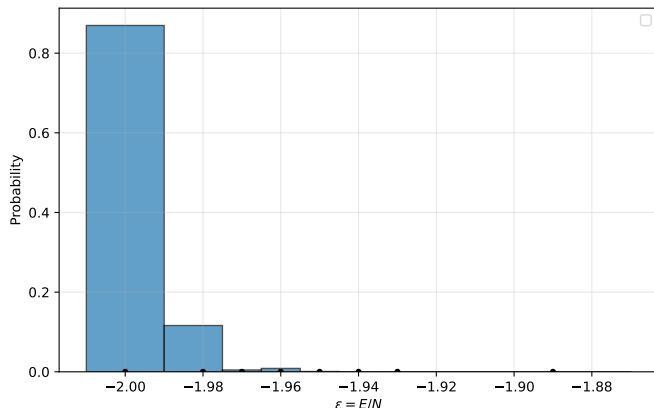


FIG. 2. Normalised histogram of ϵ for $L = 20$ at $T = 1.0 \text{ J/k}_B$ (after 100 MCC burn-in).

At $T = 1.0 \text{ J/k}_B$ the distribution collapses into a very narrow peak with most probability concentrated in one (or a few) adjacent energy levels. This reflects a nearly ordered ferromagnetic state with rare excitations; correspondingly, the variance is extremely small. By contrast, at $T = 2.4 \text{ J/k}_B$ the histogram spreads over many more energy levels with noticeably lower peak height and extended tails, indicating a much larger variance. Methodologically, using long runs with temperature-dependent burn-in ensured that the histograms are insensitive to initial conditions.

Parallel performance with OpenMP

To investigate the benefits of parallelisation, we repeated our Monte Carlo simulations for a fixed lattice size $L = 20$ while scan-

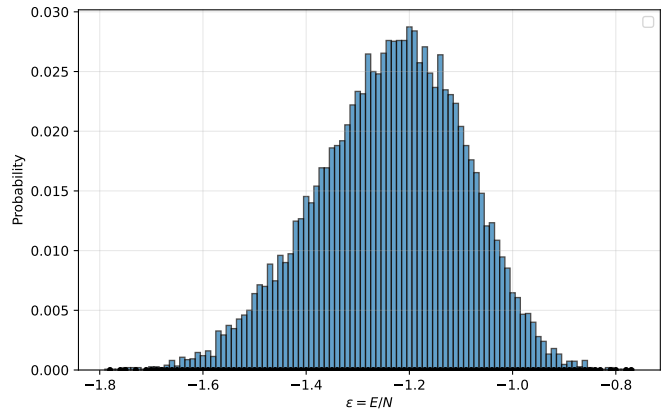


FIG. 3. Normalised histogram of ϵ for $L = 20$ at $T = 2.4 \text{ J/k}_B$ (after 10^5 MCC burn-in).

ning over a small set of temperatures. For each temperature we used $N_{MC} = 10^5$ Monte Carlo sweeps and a burn-in of 10^3 sweeps. The chosen temperatures were

$$T \in \{1.0, 2.4, 3.5, 4.6, 5.7\} \text{ J/k}_B.$$

From a computational point of view, simulations at different temperatures are independent and therefore parallel. We exploited this by parallelising the outer loop over temperatures using OpenMP, where each thread initialises its own lattice and runs the simulation at the assigned temperature. For comparison, we also executed the same set of five simulations sequentially, using a standard serial for loop over the temperature array.

The measured wall-clock times were

$$t_{\text{parallel}} = 1.47087 \text{ s}, \quad t_{\text{serial}} = 5.05582 \text{ s},$$

which yields a speed-up factor

$$S = \frac{t_{\text{serial}}}{t_{\text{parallel}}} \approx 3.44.$$

This demonstrates that, for our workload, OpenMP parallelisation over temperatures reduces the total time-to-solution. The gain is substantial and indicates that, once the Monte Carlo kernel is implemented efficiently, parallel execution across temperatures is an effective strategy for exploring parameter space in the Ising model.

Critical region

To characterise the phase transition and its finite-size effects, we extended our OpenMP-parallel simulations to larger lattices and to a dense set of temperatures around the expected critical point. For each lattice size $L \in \{40, 60, 80, 100\}$ we simulated the 2D Ising model with periodic boundary conditions for $N_{MC} = 10^5$ Monte Carlo sweeps per temperature and a burn-in of 10^3 sweeps. Temperatures were chosen in the range

$$2.1 \leq T(\text{J/k}_B) \leq 2.4,$$

with a finer spacing near $T \approx 2.27 \text{ J/k}_B$ in order to resolve the critical region more accurately. The outer loop over temperatures was parallelised with OpenMP, reusing the strategy described before, so that all temperatures for a given L were simulated concurrently.

Figures 4–7 summarise the thermodynamic observables obtained from these runs: the mean energy per spin $\langle \epsilon \rangle$, the absolute magnetisation per spin $\langle |m| \rangle$, the heat capacity per spin C_V/N and the susceptibility per spin χ/N .

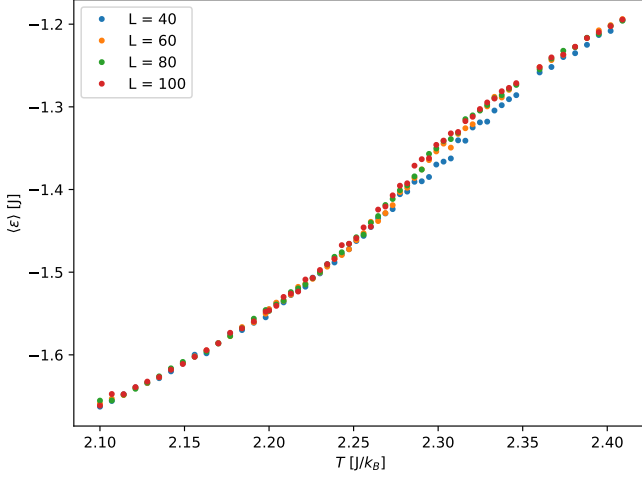


FIG. 4. Energy per spin $\langle \epsilon \rangle$ as a function of temperature for $L = 40, 60, 80, 100$.

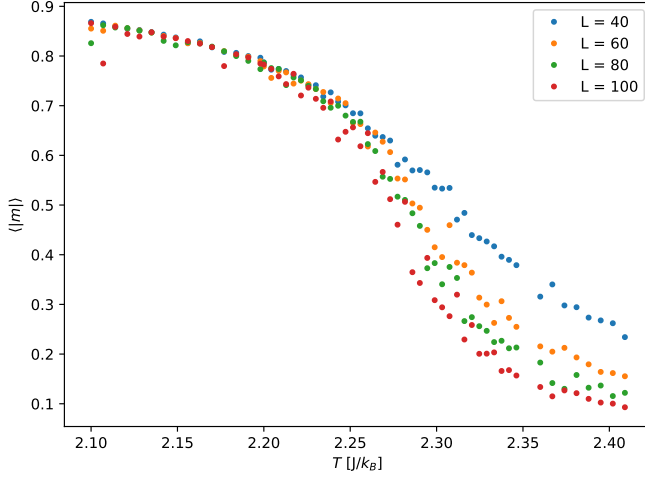


FIG. 5. Absolute magnetisation per spin $\langle |m| \rangle$ versus temperature for $L = 40, 60, 80, 100$.

The energy curves in Fig. 4 remain smooth across the whole temperature range, but their slope becomes noticeably steeper in a narrow interval around $T \simeq 2.25\text{--}2.30 J/k_B$, and this steepening is more pronounced for larger L . This change of curvature signals the underlying non-analyticity that would appear in the thermodynamic limit. The magnetisation plots in Fig. 5 show a much sharper qualitative change: for all lattice sizes the system is strongly magnetised at low temperatures and $\langle |m| \rangle$ drops rapidly to small values in the same temperature window. As L increases, the crossover in $\langle |m| \rangle$ becomes steeper and confined to an increasingly narrow interval, consistent with a genuine phase transition emerging as $L \rightarrow \infty$.

Both C_V/N and χ/N exhibit clear (marked) peaks in Figs. 6 and 7, respectively. The height of these peaks grows markedly with system size. From the heat capacity and susceptibility plots, the temperatures corresponding to the maxima of each curve can be extracted. These temperature values (which represent the critical temperature T_c for each lattice size) are reported in Table III. To obtain the values of T_c in Table III we compute the mean values of the maximums marked in Figs. 6 and 7.

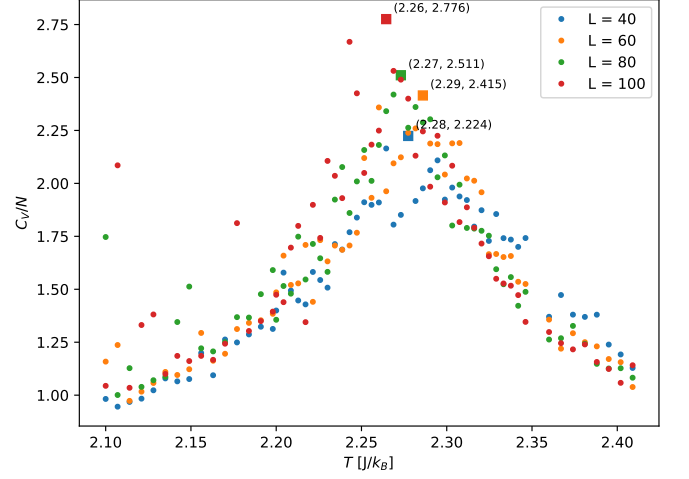


FIG. 6. Heat capacity per spin C_V/N as a function of temperature for $L = 40, 60, 80, 100$. The maximum C_V/N values are marked with colored squares.

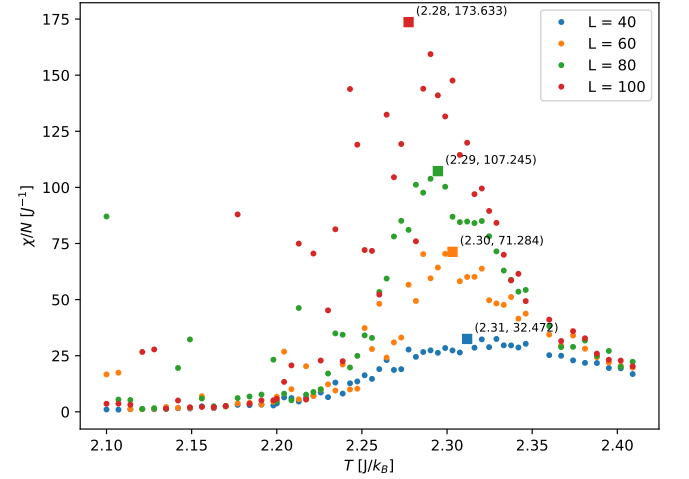


FIG. 7. Susceptibility per spin χ/N as a function of temperature for $L = 40, 60, 80, 100$. The maximum χ/N values are marked with colored squares.

TABLE III. T_c for different lattice size L .

L	$T_c [J/k_B]$
40	(2.30 ± 0.02)
60	(2.295 ± 0.005)
80	(2.28 ± 0.01)
100	(2.27 ± 0.01)

Overall, these results show that the parallel OpenMP implementation allows us to efficiently map out the temperature dependence of the Ising model for several lattice sizes, revealing a well-defined phase transition.

Estimation of $T_c(L \rightarrow \infty)$ from finite-size data

In order to estimate the critical temperature of the infinite system, we used the finite-size scaling relation

$$T_c(L) - T_c(L = \infty) = aL^{-1}, \quad (29)$$

derived in Sec. II. The finite-lattice pseudo-critical temperatures $T_c(L)$, obtained from the maxima of C_V/N and χ/N , are summarised in Table III. Using these values as input, we performed a weighted linear regression of $T_c(L)$ as a function of $1/L$, with the inverse error bars from Table III as weights. The resulting fit is shown in Fig. 8, where the data points from Table III are plotted together with the best-fit line.

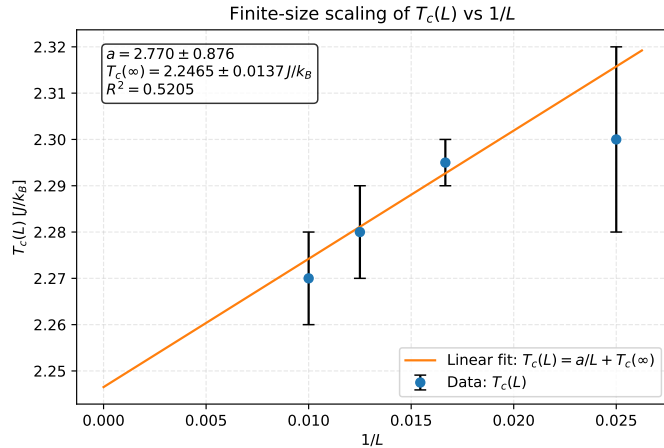


FIG. 8. Finite-size scaling of $T_c(L)$ versus $1/L$ using the values from Table III. The solid line shows a weighted linear fit $T_c(L) = a/L + T_c(\infty)$.

From the regression we obtain

$$a = 2.77 \pm 0.88, \quad T_c(L = \infty) = 2.25 \pm 0.02 \text{ J/k}_B, \quad (30)$$

with a coefficient of determination $R^2 \simeq 0.52$. The extrapolated value for $T_c(L = \infty)$ overlaps within errors with Onsager's exact result $T_c = 2.269 \text{ J/k}_B$. Given the small number of lattice sizes used and the relatively large statistical and systematic uncertainties in locating the finite-size peaks, this level of agreement is acceptable.

IV. CONCLUSION

In this work we have carried out a systematic Monte Carlo study of the two-dimensional Ising model with periodic boundary conditions, combining analytical benchmarks, numerical simulations and finite-size scaling. Starting from the exactly solvable 2×2 lattice, we showed that our C++ implementation reproduces the closed-form expressions for the energy, magnetisation, heat capacity and susceptibility at $T = 1.0 \text{ J/k}_B$ to within numerical precision. This validation step confirmed that the energy convention, periodic boundary conditions and Metropolis acceptance rule were implemented consistently.

For a 20×20 lattice we analysed equilibration from both ordered and disordered initial states. The time-series of the energy per spin revealed a very short burn-in at low temperature and a much longer approach to equilibrium near $T = 2.4 \text{ J/k}_B$, illustrating critical slowing down as the critical temperature region is approached. Using long equilibrated runs we estimated the full probability distribution of the energy per spin, finding a narrow, nearly degenerate distribution deep in the ordered phase and a broadened distribution with enhanced variance close to criticality.

We then exploited OpenMP to parallelise independent simulations at multiple temperatures. For $L = 20$ and 10^5 Monte Carlo sweeps per temperature, parallelisation of the outer temperature loop yielded a speed-up factor of about 3.4 compared to a serial run, demonstrating that shared-memory parallelism is an efficient way to reduce the time-to-solution for parameter scans in lattice models. Extending the simulations to larger lattices with $L = 40, 60, 80, 100$ and temperatures in the range $2.1\text{--}2.4 \text{ J/k}_B$, we mapped out the temperature dependence of the energy, magnetisation, heat capacity and susceptibility. The data clearly exhibit a phase transition: the magnetisation drops sharply, while both C_V/N and χ/N develop increasingly sharp peaks as L grows.

From the positions of these peaks we extracted pseudo-critical temperatures $T_c(L)$ and used the finite-size scaling relation $T_c(L) = T_c(\infty) + a/L$ to extrapolate to the thermodynamic limit. A weighted linear fit of $T_c(L)$ versus $1/L$ gave $T_c(\infty) = 2.25 \pm 0.02 \text{ J/k}_B$, which is compatible within errors with Onsager's exact result $T_c = 2.269 \text{ J/k}_B$. While the precision of this estimate is limited by the relatively small set of lattice sizes and by the resolution of the temperature grid, the agreement confirms the validity of the scaling approach and the robustness of our numerical implementation. Possible extensions of this work include a more systematic finite-size scaling analysis with additional lattice sizes, and the study of other observables or boundary conditions within the same computational framework.

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